

KARISHMA ELSA JOHNS

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EDUCATION

MASTER OF SCIENCE, Computer Science
Arizona State University, **GPA: 4.0/4.0**

May 2025
Tempe, Arizona

BACHELOR OF TECHNOLOGY, Computer Science and Engineering
APJ Abdul Kalam Technological University, **GPA: 9.34/10.0**

August 2019
Kerala, India

EXPERIENCE

Senior Applications Engineer

June 2019 - July 2023

ORACLE

Trivandrum, India

- Developed and integrated new features within the Oracle Cloud ERP Billing suite, significantly elevating system performance and enhancing the overall user experience.
- Successfully resolved over 150 critical internal and customer-reported bugs and logged over 130 incidents.
- Interacted directly with customers to troubleshoot and resolve issues, ensuring timely support and enhancing customer satisfaction.
- Optimized core functionalities through strategic code enhancements and SQL refinements, resulting in measurable performance boosts of around 40% and enhanced operational efficiency.
- Introduced over 100 automated test cases, expanding testing coverage and reducing manual testing time, leading to faster deployment cycles.
- Migrated the ERP application from Oracle 11g to 12c, ensuring that all the functionalities were intact, and modified the sanity test framework to enforce data integrity and system stability post-migration.
- Enhanced the Invoice Approval Cycle by implementing a REST API-based architecture and validated using Postman.
- Led regular code review sessions for over 50 bugs and feature enhancements.
- Led and mentored a team of three junior engineers.
- Tech stack: Java, ADF, Oracle DB, PL/SQL, Rest APIs, JavaScript, OCI, ADE, Junit, Git, XML, AngularJS, Selenium, Docker, Postman, Agile methodology

Research Intern, Tcs Lab

May 2018 - July 2018

INDIAN INSTITUTE OF TECHNOLOGY(IIT)

Chennai, India

- Collaborated with Professor C. Pandu Rangan at IIT Madras on designing and developing innovative Lightweight Proxy Re-encryption schemes for use in devices with limited computing power.
- Developed custom Python tools and scripts to automate and support experimental research, conducted in-depth research for two months, contributing to the design and implementation of secure and efficient re-encryption algorithms.
- Presented research findings to peers and faculty during weekly sync-up meetings.

SKILLS

- Languages:** C, C++, Java, Python, R, JavaScript, SQL, PL/SQL, HTML, CSS, XML
- Frameworks & Libraries:** ADF, JUnit, AngularJS, Selenium, TensorFlow, PyTorch, Keras, scikit-learn, Numpy, Pandas
- Tools & Technologies:** Oracle Cloud Infrastructure, Docker, Git, JIRA, ADE, REST APIs, Kafka, Hadoop, Postman, ROS, Spark, Kubernetes, Minikube
- Databases:** PostgreSQL, Oracle Database, MySQL, neo4j, rocksDB
- AI/ML:** Natural Language Processing (NLP), Deep Learning, Reinforcement Learning, Neural Networks, CNN, RNN, FAISS, BM25

PROJECTS & PUBLICATIONS

- Publications:** <https://scholar.google.com/citations?user=ym-fao0AAAAJ&hl=en&oi=ao>
- Code RAG** | Designed a scalable system for Python and C++ code snippet indexing, retrieval, and description and comment generation using BM25, FAISS, RocksDB, and Qwen2.5-Coder-1.5B LLM. Skills: Python, Information Retrieval, Vector Search, RocksDB, LLM Integration, Query Optimization, Scalable Data Processing, Unit Testing & Validation.
- Distributed Real-Time Graph Processing Pipeline** | Built a scalable data processing system using Kubernetes, Kafka, and Neo4j for real-time graph analytics on the NYC Yellow Cab dataset. Implemented PageRank and BFS algorithms with Neo4j GDS library in distributed environments. Skills: Kubernetes, Kafka, Neo4j, Docker, Minikube, Python, Distributed Systems, Graph Analytics, Software Integration.
- xv6 Operating System Implementation** | Built core components of the xv6 OS, such as the bootloader, linker, scheduler, page table handler (with page replacement algorithms), trap handler, multithreading, and custom virtual machine emulator. Skills: Operating Systems, C Programming, Assembly Language, System-Level Programming, Memory Management, Multithreading.
- Personally Identifiable Information (PII) Detection and Masking from Text** | Developed PII detection system using Transformer embeddings, Decision Trees, and Random Forest models to mask sensitive information. Skills: NLP, Data Privacy, Transformers, Python.
- Lightweight Stereo Image Depth Estimation** | Utilized U-Net architecture and Spatial Propagation Network to enhance depth estimation accuracy in stereo images. Skills: Computer Vision, Deep Learning, Image Processing, Python.
- Interactive Music Management System** | Built an interactive music management application using Java (Swing Library) and Oracle DB for efficient music library organization. Skills: Java, UI/UX Design, Database Management, Oracle SQL.